METRIC

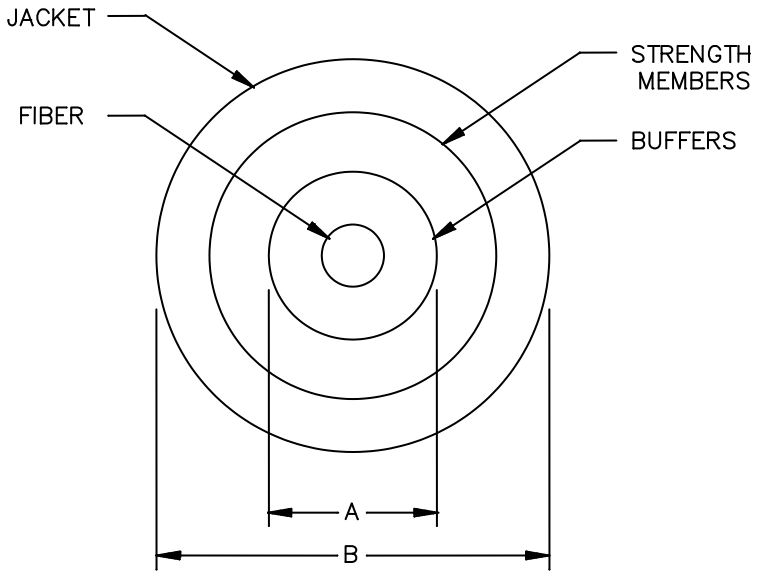
MIL-PRF-85045/31
17 September 2009

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, ONE FIBER, CABLE CONFIGURATION
TYPE 1 (BUFFERED FIBER), TIGHT BUFFER, CABLE CLASS SM AND MM,
ENHANCED PERFORMANCE CHARACTERISTICS (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein
shall consist of this specification sheet and MIL-PRF-85045.



µm	Inches
900	.035
50	.002

mm	Inches
2.2	.087
1.8	.070

Dimensions	
A µm	B mm
900 ± 50	1.8 – 2.2

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for reference purposes.

FIGURE 1. Tight buffered single fiber optical cable.

AMSC N/A

FSC 6015

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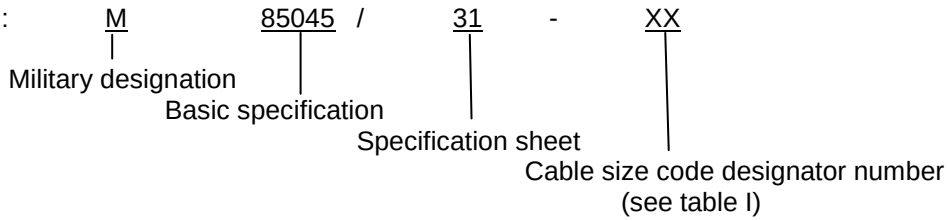
CLASSIFICATION:

Fiber optic cable configuration type: 1 (Buffered fiber).

Fiber cable class: MM (Graded-index, glass core and glass cladding, multimode).
SM (Dispersion-unshifted, glass core and glass cladding, single-mode).

Part or Identifying Number (PIN): See PIN construction below. Additional manufacturer's marking is allowed.

PIN construction:



PIN example: M85045/31-01 is a single mode cable consisting of enhanced 9/125 micron fiber with an operating temperature range -55° C to +165° C, and fiber coating of 125 micron

TABLE I. Cable code designator letter.

Cable size code designation number	Fiber size (microns)	Fiber size code (see MIL-PRF-49291/12)	Operating temperature range °C (°F)	Coating diameter (microns)
01	9/125	01	-55 to +165 (-67 to 329)	125
03	9/125	02	-55 to +165 (-67 to 329)	250
05	9/125	01A	-40 to +85 (-40 to 185)	125
07	9/125	02A	-40 to +85 (-40 to 185)	250
02	62.5/125	01	-55 to +165 (-67 to 329)	125
04	62.5/125	02	-55 to +165 (-67 to 329)	250
06	62.5/125	01A	-40 to +85 (-40 to 185)	125
08	62.5/125	02A	-40 to +85 (-40 to 185)	250
20	50/125	01	-55 to +165 (-67 to 329)	500
40	50/125	02	-55 to +165 (-67 to 329)	245
60	50/125	01A	-40 to +85 (-40 to 185)	500
80	50/125	02A	-40 to +85 (-40 to 185)	245

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DESIGN AND CONSTRUCTION:

Fiber:

Class MM 62.5/125 micron fibers shall be in accordance with MIL-PRF-49291/10.
Class SM enhanced 9/125 micron fibers shall be in accordance with MIL-PRF-49291/11.
Class MM 50/125 micron fibers shall be in accordance with MIL-PRF-49291/12.

Buffer diameter: 900 ± 50 µm.

REQUIREMENTS:

Finished cable:

Dimensions and configuration: See figure 1.

Mass per unit length: $\leq$ 5.0 kg/km.

Color: Slate for class MM fiber with a 62.5/125 micron core/cladding diameter.
White for class MM fiber with a 50/125 micron core/cladding diameter.
Yellow for class SM fiber with an enhanced 9/125 mode field diameter.
Orange for cable used in flight test application only, regardless of fiber size.

Materials:

Jacket material: The jacket shall be composed of a low halogen, low smoke, low toxicity polymer material.

Fill: Jacket shall not contain fill material.

Jacket red light transparency: Fiber breaks inside the cable jacket shall be easily recognized when pulsed, red light from a laser (monochromatic light) is sent through the optical fiber in the cable. The monochromatic light shall have a nominal wavelength of 635 nm, a nominal pulse rate of 3 Hz and a minimum optical power level of 1 mW.

Number of fibers: 1.

Concentricity: $\leq$ 0.8.

Short term minimum bend diameter: Eight times the cable outer diameter. (The short term minimum bend diameter is to be used in all environmental and mechanical tests which specify a cable minimum bend diameter.)

Long term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, quality conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

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PERFORMANCE REQUIREMENTS:

Qualification inspection. Test performed, test sequence, test procedure, cable performance and test sample (specimen) quantity and length shall be in accordance with table II with any exception (such as a modification or addition) listed in this specification sheet.

TABLE II. Qualification inspection.

Group	Inspection	Requirement paragraph	Test paragraph	Specimen length 1/ 2/
I	Visual & mechanical inspection	3.4,3.9,3.10	4.7.2	3 units, 0.5 km each 3/
	Attenuation rate	3.5.1	4.7.4.1	3 units, 0.5 km each 4/
II	Crosstalk	3.5.2	4.7.4.3	3 units, 6 m each 4/
	Electromagnetic effects	3.7.16	4.7.6.16	1 unit, 6 m 5/
III	Temperature cycling	3.7.1	6/	2 units, 0.5 km each 4/ (1 on reel, 1 off)
	Temperature cycling with mandrel	3.7.1	4.7.6.1	2 units, 0.5 km each 4/ (1 on reel, 1 off)
	Temperature humidity cycling	3.7.3	4.7.6.3	2 units, 0.5 km each 7/
	Storage temperature	3.7.4	4.7.6.4	2 units, 0.5 km each 7/
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	2 units, 8 m each 8/
	Cyclic flexing	3.6.4	4.7.5.4	6 units, 5 m each 9/ (2 units at each temp)
	Crush	3.6.5	4.7.5.5	2 units, 5 m each 8/
	Cable twist-bend	3.6.6	4.7.5.6	6 units, 5 m each 9/ (2 units at each temp)
	Impact	3.6.8	4.7.5.8	2 units, 5 m each 8/
	Barometric pressure (reduced)	3.7.5	4.7.6.5	1 unit, 0.46 km each 10/
	Life aging (temperature life)	6/	6/	2 units, 300 m each 8/
	Tensile loading and elongation	3.6.1	4.7.5.1	2 units, 150 m each 11/
	Operating tensile loading	3.6.1.1	4.7.5.1.1	2 units, 150 m each 12/
	Freezing water immersion (ice crush)	3.7.7	4.7.6.7	2 units, 30 m each 8/
	Fungus resistance	3.8.4	4.8.4	2 units, 0.5 m each 8/
	Cable element removability	3.6.18	4.7.5.18	2 units, 0.5 m each 8/
	Flammability (60 degree angle)	3.7.12.1	4.7.6.12.1	2 units, 6 m each 8/

See footnotes at end of table

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TABLE II. Qualification inspection - Continued.

Group	Inspection	Requirement paragraph	Test paragraph	Specimen length 1/ 2/
IV	Thermal shock	3.7.2	4.7.6.2	1 unit, 0.49 km each 4/ (on reel)
	Fluid immersion	3.7.8	4.7.6.8	1 unit, 1 m 13/ and 3 material samples 14/ for each specified fluid
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	1 unit, 30 m 13/
	Vibration	6/	6/	1 unit, 30 m 13/
	Shock	3.7.13	4.7.6.13	1 unit, 30 m 13/
	Dynamic bend	3.6.2	4.7.5.2	1 unit, 150 m 15/
	Corner bend	3.6.9	4.7.5.9	2 units, 5 m 16/
	OFCC kink	3.4.2.2.2	4.7.3.5	2 units, 1 m 13/
V	Dripping	3.6.13	4.7.5.13	1 unit, 30 cm 13/
	Cable jacket material tensile strength & elongation	3.6.15	4.7.5.15	5 units 17/
	Cable abrasion resistance, cable-to-cable	3.6.16	4.7.5.16.2	4 units, 2 m 18/
	Cable shrinkage	3.6.17	4.7.5.17	3 units, 0.5 m 16/
	Durability of identification	3.6.19	4.7.5.19	3 units, 2 m each 16/
	Flaming smoke generation	6/	6/	1 unit 19/
	Continuous flame	6/	6/	1 unit 20/
	Wicking	3.7.10	4.7.6.10	2 units, 2 m 16/
	Cable strain relief pull out force	6/	6/	5 units, 2 m each 16/
VI	Acid gas generation	3.8.1	4.8.1	1 unit, 1 m 21/
	Halogen content	3.8.2	4.8.2	1 unit, 1 m 21/
	Toxicity index	3.8.3	4.8.3	1 unit, 1 m 21/
	Outgassing, thermal vacuum	3.3.6.1	4.8.5.1	1 unit, 1 m 21/

1/ Tolerance on 0.5 km length is ± 5 percent provided results are normalized to 1 km.
2/ Tolerance on shorter lengths is ± 5 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2 m section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ A specimen cut from one of the visual and mechanical inspection test samples shall be used.
6/ As stated under the applicable tests in this specification sheet.
7/ The same samples as used in the temperature cycling test shall be used.
8/ A specimen cut from each sample used in the temperature cycling test shall be used.
9/ Three specimens cut from each sample used in the temperature cycling test shall be used.
10/ One of the samples used in the temperature cycling test shall be used.

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TABLE II. Qualification inspection - Continued.

11/ A specimen cut from each sample used in the life test shall be used.
12/ The same sample as used in the tensile loading and elongation test shall be used.
13/ A specimen cut from the sample used in the thermal shock test shall be used.
14/ Three jacket material samples as specified in 4.7.5.15 shall be used in this test.
15/ A specimen cut from the sample used in the thermal shock test or one of the samples used in the tensile loading and elongation test shall be used.
16/ Specimens cut from the sample used in the thermal shock test shall be used.
17/ Jacket material samples as specified in 4.7.5.15 shall be used in this test.
18/ Specimen cut from the sample used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion and two specimens shall be used in the cable to cable abrasion testing.
19/ Use 76 mm (3 inch) lengths of cable adjacent to one another to form a 76 mm (3 inch) strip. For a 2 mm (.08 inch) diameter cable, an approximate length of 3 m is needed.
20/ Use 305 mm (12 inch) lengths of cable adjacent to one another to form a 457 mm (18 inch) strip. For a 2 mm (.08 inch) diameter cable, an approximate length of 69 m is needed.
21/ A specimen cut from one of the samples used in the temperature cycling or thermal shock tests shall be used.

Preconditioning. No preconditioning shall be performed as part of qualification. If preconditioning is required, it shall be performed as part of cable fabrication process and supplied already preconditioned.

Optical properties:

Maximum attenuation rate:

For cable with MM 62.5/125 micron fiber: 4.0 dB/km at 850 nm ± 20 nm, 2.0 dB/km at 1,300 nm ± 20 nm, and otherwise 4.5 dB/km over the wavelength range from 800 nm to 1600 nm.

For cable with SM 09/125 micron fiber: 1.0 dB/km at 1,310 nm ± 20 nm, 1.0 dB/km at 1,383 nm ± 20 nm, 0.75 dB/km at 1,550 nm ± 20 nm, 0.75 dB/km at 1,625 nm ± 20 nm, and otherwise 1.0 dB/km over the wavelength range from 1,250 nm to 1,650 nm.

For cable with MM 50/125 micron fiber: 4.0 dB/km at 850 ± 20 nm, 2.0 dB/km at 1,300 nm ± 20 nm, and otherwise 4.0 dB/ km at 1,300 nm ± 20 nm, and otherwise 4.5 dB/km over the wavelength range from 800 nm to 1600 nm.

Bandwidth (multimode cables only):

For cable with MM 62.5/125 micron fiber: $\geq$ 200 MHz-km at 850 nm ± 20 nm, $\geq$ 500 MHz-km at 1,300 nm ± 20 nm for cable with MM fiber

For cable with MM 50/125 micron fiber: $\geq$ 1500 MHz-km at 850 nm ± 20 nm, $\geq$ 500 MHz-km at 1,300 nm ± 20 nm for cable with MM fiber

Polarization mode dispersion (single mode only): $\leq$ 0.2 ps/$\sqrt{\ }$km

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Change in optical transmittance: Measurements to be made at 850 nm ± 20 nm for multimode and 1550 ± 20 nm for single mode.

Crosstalk: Applicable except each specimen length shall be 6 m when testing three adjacent single fiber cables. The middle portions of the three adjacent cables shall be wrapped around a 15 ± .5 mm (.59 ± .02 inch) mandrel for 5 turns.

Mechanical properties:

Tensile loading and elongation: Applicable except tensile loading ≥ 270 N.

Operating tensile loading: Applicable.

Dynamic bend: Applicable except tensile load ≥ 90 N.

Cyclic flexing: 500 cycles at 25° C ± 2° C and 100 cycles at -55° C ± 2° C (or -40 ° C ± 2° C for cable specified with an operating temperature of -40° C to 85° C). Change in optical transmittance measurements are to be made every 100 cycles for the 500 cycle exposure and every 25 cycles for the 100 cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Crush: Applicable except that the compression loading shall not be less than 500 N/cm times the outer cable jacket in cm, shall be held for 1 minute and released and shall be applied at a rate of not less than 250 N/min.

Cable twist bending: 500 cycles at 25°C ± 2°C and 100 cycles at -55°C ± 2°C (or -40°C ± 2°C for cable specified with an operating temperature of -40°C to 85°C). The test load shall be 10 N. Change in optical transmittance measurements are to be made every 100 cycles for the 500 cycle exposure and every 25 cycles for the 100 cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Radial compression: Not applicable.

Impact: Applicable except test shall be performed for 50 cycles at 25°C ±2°C and for 20 cycles at the minimum operating temperature.

Barometric pressure (reduced): Applicable except that the change in optical transmittance shall be performed during and after the test.

Hosing: Not applicable.

Dripping: Applicable except test is to be performed at 200 ºC (392 ºF).

Environmental properties:

Temperature range: See table III.

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TABLE III. Temperature range.

PIN 1/	Operating °C (°F)	Nonoperating °C (°F)	Storage °C (°F)
M85045/31-01,-03, for 09/125 -02,-04, for 62.5/125 2/ -20,-40, for 50/125	-55 to +165 (-67 to 329)	-40 to + 85 (-40 to 185)	-40 to +85 (-40 to 185)
M85045/31-05,-07, for 09/125 -06,-08, for 62.5/125 3/ -60,-80, for 50/125	-40 to +85 (-40 to 185)	-55 to +85 (-67 to 185)	-55 to +85 (-67 to 185)

1/ See table I for PIN fiber size code designator number.
2/ Temperature ranges in row 2 conform to those specified for other MIL-SPEC fiber optic cabling components for aircraft applications.
3/ Temperature ranges in row 3 conform to those specified for other MIL-SPEC fiber optic cabling components for standard applications (such as the telecommunication industry).

Temperature cycling: Applicable except perform the test as follows for the operating temperature range of -55°C to +165°C. Optical fiber shall be tested in accordance with TIA-455-3 using the number of cycles, test condition schedule and soak times in accordance with the table IV. The change in optical transmittance shall be measured during and after the test. At a minimum for the "during test" measurements, an optical transmittance measurement shall be performed towards the end of each soak period (maintain step in table IV) after every 50 cycles. A post test visual examination of the test specimens shall reveal no surface impairment or any damage detrimental to the operation of the test specimens. The operating temperature range shall be as specified herein. Test fixtures, if used must be of minimum mass and approved by the qualifying activity. No other mass (item that causes significant thermal lag) shall be added inside the chamber.

TABLE IV Temperature cycling steps.

Step	Cycle	Action	Temperature °C (°F)	Duration 1/
1	1	Maintain	25 +/-2 (77 ±4)	4 hours (minimum)
2		Ramp to	-55 +0/-3 (-67 +0/-5)	8 minutes
3		Maintain	-55 +0/-3 (-67 +0/-5)	15 minutes (minimum)
4		Ramp to	+165 +3/-0 (392 +5/0)	22 minutes
5		Maintain	+165 +3/-0 (392 +5/-0)	15 minutes (minimum)
6	2	Ramp to	-55 +0/-3 (-67 +0/-5)	22 minutes
7		Maintain	-55 +0/-3 (-67 +0/-5)	15 minutes (minimum)
8		Ramp to	+165 +3/-0 (392 +5/0)	22 minutes
9		Maintain	+165 +3/-0 (392 +5/-0)	15 minutes (minimum)
10	3-1,000	Repeat steps 6 thru 9, 998 additional times, for a total of 1,000 cycles.		
11	Post 1,000	Ramp to	25 +/-2 (77 ±4)	6 minutes
12		Maintain	25 +/-2 (77 ±4)	4 hours (minimum)

1/ Ramp rate is 10°C per minute, soak time (maintain step) is 15 minutes.

Temperature cycling with mandrel: Applicable except perform with portion of cable being wrapped around a 20 ±1 mm (.79 ±.04 inch) mandrel for 5 turns.

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Temperature humidity cycling: Applicable except that the change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Temperature life: Applicable except perform as follows for the operating temperature range of -55°C to +165°C. The high exposure temperature shall be 165°C +5°C, -0°C. Test samples shall be tested in accordance with TIA/EIA-455-4 for the duration of 1,000 hours at the high exposure temperature. The change in optical transmittance shall be measured after the test. The optical fiber shall be visually examined after the test to the extent feasible. Fiber Optic cable shall not be damaged, and there shall be no distortion, leaking of coating or other damage detrimental to the operation of the optical fiber. The requirement for the change in optical transmittance shall be met after the test. Performance of a cable/fiber retention test is not required as part of the final inspection for this test.

Storage temperature: Applicable.

Weathering: Not applicable.

Fluid immersion: Applicable except that specimens shall be exposed to the all of the fluids at the temperatures specified in table V. Specimens shall be maintained at ambient conditions for a minimum of 4 hours prior to fluid testing. After immersion, all specimens shall be drained (for at least 10 minutes), blotted to remove excess fluid (without scrubbing or mechanically interact), rinsed with moderate agitation (using only a back and forth or up and down type motion) for a maximum of 30 seconds in isopropyl alcohol (TT-I-735), blotted dry (without scrubbing or mechanically interact), and air dried for a minimum of one hour prior to inspection and transfer to subsequent fluid. After testing, each specimen shall be visually examined for conformance to the inspection requirements.

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TABLE V. Test fluids.

Category	Fluid type	Specification 1/	Test temperature °C (°F)
Fuel	Turbine engine fuel reference fluid	SAE AMS2629, Type 1 2/	60 (140)
	Fuel	MIL-DTL-16884	35 (95)
	Automobile gasoline	ASTM D 4814	25 (77)
Hydraulic fluid	Petroleum based, red	MIL-PRF-5606	70 (158)
	Synthetic hydrocarbon/diester based	MIL-PRF-83282	70 (158)
	Synthetic hydrocarbon /PAO based, red	MIL-PRF-87257	70 (158)
	Petroleum, inhibited	MIL-PRF-17672	60 (140)
	Phosphate ester based	SAE AS1241, Ty IV, Cl 1	70 (158)
Lubricating oil	Moderate service	MIL-PRF-17331	121 (250)
	Synthetic base	MIL-PRF-23699	121 (250)
Coolant fluid	Coolanol 25R	3/	60 (140)
	PAO based dielectric fluid	MIL-PRF-87252	60 (140)
Deicer	Aircraft	SAE AMS1424	60 (140)
	Runway	SAE AMS1435	60 (140)
Water	Sea	ASTM D 1141	60 (140)
	Reagent 4/	ASTM D 1193	60 (140)
Cleaners/solvents	Isopropyl alcohol	TT-I-735 5/	60 (140)
	Methyl Isobutyl Ketone	ASTM D 1153	60 (140)
	Alkaline detergent	MIL-PRF-85570 6/	60 (140)
Corrosion preventive compound		MIL-PRF-16173, Gr 4	60 (140)
		MIL-PRF-81309, Ty III	60 (140)

1/ Test duration for all fluids is 48 hours.
2/ JP-8 conforming to MIL-DTL-83133 may be used in lieu of SAE AMS2629 only if the aromatic content is increased to 25 percent using toluene.
3/ Source of supply: Exxon-Mobile Chemical
4/ Reagent water is specified in lieu of tap water.
5/ Reagent grade only.
6/ Type II tested with a 4:1 dilution.

Flammability (60 degree angle): Applicable.

Flame extinguishing: Not applicable.

Smoke generation and flame propagation: Not applicable.

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Flaming smoke generation: $D_m \leq 225$ when tested in the flaming condition in accordance with ASTM-E-662. The size and configuration of the test specimen shall conform to the 76 mm (3 inch) by 76 mm (3 inch) square specified in ASTM-E-662. The thickness of the test specimen shall be 2 mm (.08 inch). The specimen shall be constructed by laying 3 inch lengths of cable adjacent to one another to form a 76 mm (3 inch) strip. Wire or cable clamps may be used to permanently hold the cable lengths in place. Longer cable lengths may be used if the top or bottom clamps are used to hold the cables. If top or bottom clamps are used, the length of cable exposed outside of the clamp(s) shall be 76 mm (3 inches).

Continuous flame: Flame spread factor, $F_S \leq 2.75$, Heat evolution factor, $Q \leq 6$ when performed in accordance with ASTM-E-162. The thickness of the test specimen shall be 2 mm (.08 inch). The specimen shall be constructed by laying 305 mm (12 inch) lengths of cable adjacent to one another to form a 457 mm (18 inch) strip. Wire or cable clamps may be used to permanently hold the cable lengths in place. Longer cable lengths may be used if the top or bottom clamps are used to hold the cables. If top or bottom clamps are used, the length of cable exposed outside of the clamp(s) shall be 305 mm (12 inches).

Vibration: Applicable except cabling shall be tested for random vibration in accordance with the vibration test listed below. Optical discontinuities shall be measured during the test. A minimum of four specimens shall be monitored for optical discontinuity. The change in optical transmittance shall be monitored after the test. The cabling shall be visually examined after the test. The requirement for optical discontinuity shall be met during the test and the requirement for the change in optical transmittance shall be met after the test.

Random vibration shall be performed in accordance with TIA/EIA-455-11, test condition VI tailored using the power spectral density versus frequency curve in accordance with MIL-DTL-38999, figure 25 which starts at 25 Hz. Duration shall be 8 hours in the longitudinal direction (orientation parallel to the optical fiber length and designated at the z axis) and 8 hours in the radial direction (orientation parallel to the optical fiber end cross section and in the x-y plane) for a total of 16 hours. Vibration shall be done at ambient temperature. Calculated overall Grms value from vibration machine controller shall be provided.

Shock: Applicable except specimens shall be tested in accordance with the applicable mechanical shock tests listed below. Optical discontinuities shall be measured during the test. A minimum of four specimens shall be monitored for discontinuity. The change in optical transmittance shall be monitored after the test. The cabling shall be visually examined after the test. The requirement for optical discontinuity shall be met during the test and the requirement for the change in optical transmittance shall be met after the test.

a. MIL-S-901. Test samples shall be tested in accordance with MIL-S-901, grade A, type B, class I. Standard shock fixture 4A for bulkhead mounting shall be used. Supplement test fixture with mounting shall be performed as specified in NAVSEA Drawing 8328898.

b. Half-sine pulse. Test samples shall be tested in accordance with TIA-455-14, test condition D. Three shocks in each direction shall be applied along the three mutually perpendicular axes of the test sample (18 shocks).

Corner bend. Applicable except that test force shall be 132 N.

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OFCC kink: Applicable except that cable shall not kink at diameters above the minimum short term bend diameter.

Jacket material tensile strength and elongation: Tensile strength 900 N/cm^2 minimum, elongation 125 percent minimum.

Cable shrinkage: Applicable except total shrinkage of 30 cm length shall not be greater than 3.18 mm (.125 inch).

Paint susceptibility: Not applicable.

Electromagnetic effects: Applicable.

Wicking: Applicable except maximum water penetration shall be 25 mm (1 inch).

Cable strain relief pull out force. The pull out force of the cable strain relief (that is, the cable member for strain relief to a connection device comprised of a material such as arimid yarn) shall be tested in accordance with TIA-455-6 by applying a 98 N (22 lb) force for a minimum duration of one minute. One end on each length of fiber optic cable shall be terminated onto the end of a MIL-PRF-29504 terminus (or other suitable fiber optic connector if approved by the Qualifying Activity prior to test sample fabrication). Fabrication shall be performed as specified in NAVSEA Drawing 8283460. A copy of this document can be obtained at web site: https://fiberoptics.nswc.navy.mil/ in the NAVSEA Drawing section under Component Information. Test sample shall consist of a MIL-MIL-PRF-29504/18 terminus affixed to the 2 meter cable sample at one end and a suitable connector to mate with the optical instrumentation detector interface at the other cable sample end. A second test cable shall interface the sample under test through a MIL-PRF-29504/18 terminus at one cable end and the optical source at the other cable end. An axial tensile load shall be applied on the detector end cable through a mandrel or other approved means. Load shall not be applied to the coupling mechanism (such as a MIL-PRF-64266 connector). The change in optical transmittance shall be measured during and after the test and meet the optical requirement for this measurement. The cable strain relief shall be visually inspected after the test. No evidence of strain relief damage, breakage, or disengagement (pull out) shall occur.

Chemical properties:

Halogen content: < 0.2 percent.

Outgassing, thermal vacuum: Applicable for non metallic materials only.

NOTES:

Intended use:

This cable is intended for use as cable runs between fiber optic switches (backbone cables), as cable runs to end user equipment (cable drops), jumper cordage and pigtails for fiber optic components. This cable is intended for use inside of protected enclosures (such as convoluted tubing or cable wrap)/cable harnesses and is not intended for installation in the overheads or cableways outside protective enclosures.

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Reference documents. In addition to MIL-PRF-85045, this document references the following:

MIL-S-901	MIL-PRF-29504/18	MIL-PRF-83282	SAE AMS 1435	TIA 455-3
MIL-PRF-5606	MIL-DTL-38999	MIL-PRF-85570	SAE AMS 2629	TIA-455-6
MIL-PRF-16173	MIL-PRF-49291/10	MIL-PRF-87252	ASTM E 162	TIA/EIA-455-4
MIL-DTL-16884	MIL-PRF-49291/11	MIL-PRF-87257	ASTM E 662	TIA/EIA-455-11
MIL-PRF-17331	MIL-PRF-49291/12	NAVSEA drawing 8283460	ASTM D 1141	TIA-455-14
MIL-PRF-17672	MIL-PRF-64266	NAVSEA drawing 8328898	ASTM D 1153	TT-I-735
MIL-PRF-23699	MIL-PRF-81309	SAE AS 1241	ASTM D 1193	
MIL-PRF-29504	MIL-DTL-83133	SAE AMS 1424	ASTM D 4814	

Custodians:
Army - CR
Navy - AS
Air Force - 85

NASA - NA
DLA – CC
Review activities:
Army - AR, AV, MI
Navy – SH, EC, YD
Air Force - 02, 19, 99

Preparing activity:
DLA - CC

(Project 6015-2009-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www://assist.daps.dla.mil.

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